∞-brush: Controllable Large Image Synthesis with Diffusion Models in Infinite Dimensions

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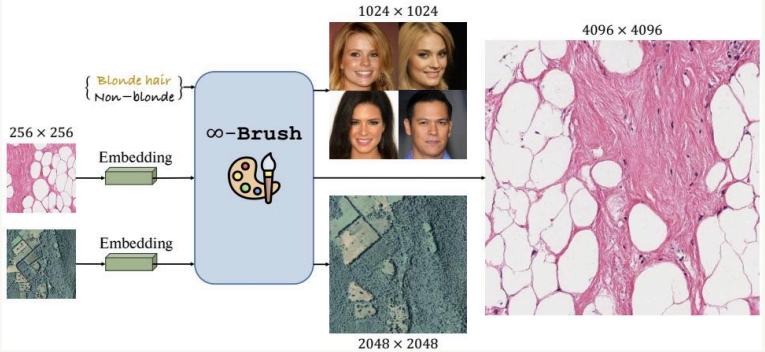


Problem Statement

- Current diffusion models limited to fixed training resolutions
- Cannot exceed 1024×1024 without fidelity loss $e.g., SDXL$
- Patch-based methods fail to capture long-range spatial relationships
- Heavy reliance on local information during generation
- No existing model enables controllable generation at 4096×4096
- Computational demands scale quadratically with resolution

Our Solution

- Introduce ∞-Brush: First conditional diffusion model in function space
- Generates images at arbitrary resolutions up to 4096×4096
- Key innovation: Cross-attention neural operator
- Enables conditioning in infinite-dimensional space
- Training efficiency: Only 0.4% pixel subsets per image
- $65,536$ pixels instead of 16 million for 4096×4096

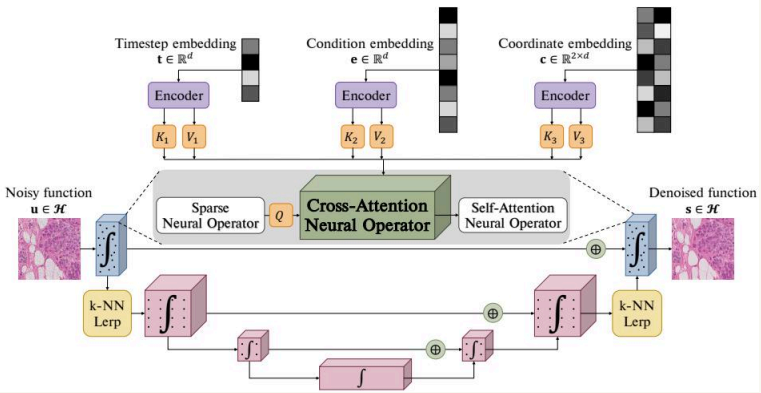


Related Work Limitations

- **Finite-dimensional models** $SDXL, Matryoshka$
- Fixed resolution constraint: Cannot exceed training resolution
- Quadratic computational cost with resolution
- **Patch-based methods** $MultiDiffusion, Graikosetal.$
- Captures local details but loses global structure
- No conditioning for controllable generation
- **Infinite-dimensional models** $\infty - Diff$
- Enables arbitrary resolution but no conditional control

Core Innovation

- **Cross-attention neural operator**
- Replaces vanilla attention in function space
- Complexity: $\mathcal{O}((N + \sum N_l)d^2)$ vs. $\mathcal{O}(N^2 d)$
- **Conditional diffusion process**
- Forward: $\mathbb{Q}(u_t|u_0) = \mathcal{N}(u_t; \sqrt{\bar{\alpha}_t}Au_0, (1-\bar{\alpha}_t)ACA^T)$
- **Training efficiency**
- 0.4% pixel subsets per image ($65,536$ vs. 16M pixels)

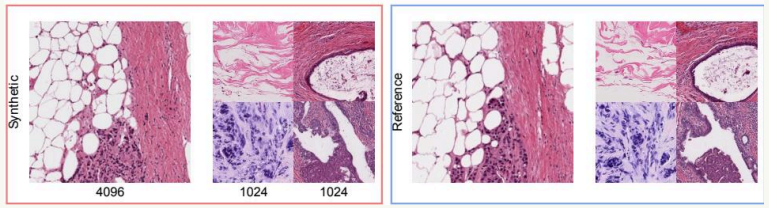


Experimental Setup

- **Datasets**
- CelebA-HQ (1024×1024): 30k images + hair color conditioning
- TCGA-BRCA (4096×4096): 57k histopathology patches
- NAIP satellite (1024×1024): 35k images
- **Evaluation metrics**
- CLIP FID: Global structure quality
- Crop FID: Local detail fidelity
- **Baselines**
- SDXL, Graikos et al. $patch-based$, ∞-Diff

Results: Global Structure

- **TCGA-BRCA** 4096×4096
- ∞-Brush: CLIP FID = 3.81 $vs. 7.43 for patch-based$
- *Takeaway:* Superior global structure preservation
- **NAIP** 1024×1024
- ∞-Brush: CLIP FID = 6.32 $vs. 10.90 for SDXL$
- **SDXL infeasible at** 4096×4096
- Requires full-resolution training $impossible on standard hardware$



Results: Efficiency & Trade-offs

- **Cross-attention operator critical**
- Ablation: Removes operator → CLIP FID ↑ 1.5, Crop FID ↑ 5.2
- **Resource comparison** $A100GPU$
- ∞-Brush: 450M params, 20 batch size, 12 hr/epoch
- SDXL: 3.5B params, 4 batch size, 140 hr/epoch
- *Takeaway:* Consistent scalability across resolutions
- **Fine detail trade-off**
- Crop FID higher than patch-based $17.87vs.15.51$
- Due to single global embedding vs. 16 local conditions

Key Contributions

- First conditional diffusion model in function space
- Enables controllable generation at arbitrary resolutions
- Cross-attention neural operator for infinite dimensions
- Linear complexity vs. quadratic for vanilla attention
- Training on 0.4% pixel subsets
- Achieves 4096×4096 synthesis with 10x lower memory



Limitations & Future Work

- **Limitations**
- Fine details slightly worse than patch-based methods
- Smaller parameter count $450Mvs.860Mfor[10]$
- **Future work**
- Design local neural operators for fine details
- Transfer knowledge from finite-dimensional models
- Scale to higher resolutions (8192×8192)

Conclusion

- ∞-Brush overcomes resolution limits of existing models
- First to enable controllable 4096×4096 synthesis
- Key innovation: Cross-attention neural operator
- Maintains global structure while preserving details
- Computational efficiency:
- 12 hr/epoch vs. 140 hr for SDXL at 1024×1024
- Constant training cost across resolutions